Sensor Fusion System for Localization of Autonomous Car

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***Abstract*— In the realm of advancing self-driving vehicle technology, numerous pivotal elements contribute significantly to its progress, with localization standing out as a critical aspect. A robust localization system holds immense significance, allowing a vehicle to precisely ascertain its position on the global map and navigate the complexities of the road network efficiently. Diverse methodologies, including the Global Navigation Satellite System (GNSS), Wheel Odometry, and Inertia Measurement, can be employed to develop this intricate system. Each method brings its own set of advantages and drawbacks, contributing to the nuanced landscape of self-driving vehicle localization. To enhance the accuracy and quality of data, this paper proposes a methodology that integrates information from various sources, such as GNSS, Wheel Odometry, and Inertia Measurement, using the Extended Kalman Filter. This approach serves as a linchpin for obtaining precise and reliable data, aiming to synergistically leverage the strengths of each method while mitigating their weaknesses. The overarching objective is to fortify the overall reliability of the generated data, emphasizing a holistic and integrated perspective. This methodology represents a significant stride toward refining data quality within self-driving vehicle localization systems, contributing to the realization of more robust and dependable autonomous driving technology.**

Keywords—GNSS, Odometry, IMU, Sensor Fusion, Extended Kalman Filter.

I. Introduction

Technological developments in the automotive world are currently encouraging unmanned vehicles which were developed into a mode of mass transportation [1]. Currently, the existence of autonomous cars as mass transportation is very realistic in industrial environments, offices, campuses, and other public spaces, where in these environments there is high mobility and the environmental conditions are safer for autonomous vehicles compared to highways, for several reasons. Among other things, road conditions are more controlled. The behavior of the surrounding environment has fewer variations compared to highways. Safety is more guaranteed if an accident occurs because the speed limit is slower than on highways in general. Localization systems can be built using various methods, including the Global Navigation Satellite System (GNSS), wheel Odometry, and Inertia Measurement Unit (IMU) [2]. Each of these methods has its own advantages and disadvantages, such as the GNSS system which has good accuracy when it is in an open condition, and the satellite signal it captures is good but has the disadvantage of low frequency and its accuracy will decrease when it is in a closed space or on a satellite signal. obtained is low. Meanwhile, in the wheel odometry system, the data update frequency is very fast, but the further the car travels, the greater the error will be. Meanwhile, the IMU system is only able to provide good data on heading data from the car, while the position data obtained will have large errors due to the nature of the IMU itself which is a combination of gyroscope, accelerometer, and compass sensors which only provide pitch, roll, and yaw rotation data, heading, and acceleration data only.

To obtain a robust localization system, a fusion algorithm is needed from various methods and sensors so that it can compensate for the weaknesses of each method and sensor use. Sensor fusion involves combining information from multiple sensors to obtain a more accurate and reliable estimate of the vehicle's state (position, orientation, velocity). The idea is that while individual sensors may have limitations or uncertainties, combining their outputs can compensate for these limitations and provide a more robust localization solution The challenge in sensor fusion is sensor calibration and the computational complexity involved. Sensor biases occur due to poorly calibrated sensors, navigational errors in moving platforms, imperfectly modeled components of target motion, and or atmospheric effects [3] and real-time processing of data from multiple sensors can be computationally intensive.

II. Methods

A. System Design

The system is created with three sensor system inputs, namely wheel odometry, GNSS, and IMU, each data will be processed to obtain data with the same units, namely the position in Cartesian coordinates of the car, after that, the two processed data will be used as input for the fusion system. In the fusion system, observers will be made regarding the quality and probability of the sensors which will be used as parameters as additional constants in the extended Kalman filter (EKF) system. After going through the EKF system, the coordinate and orientation data of the car will be obtained.

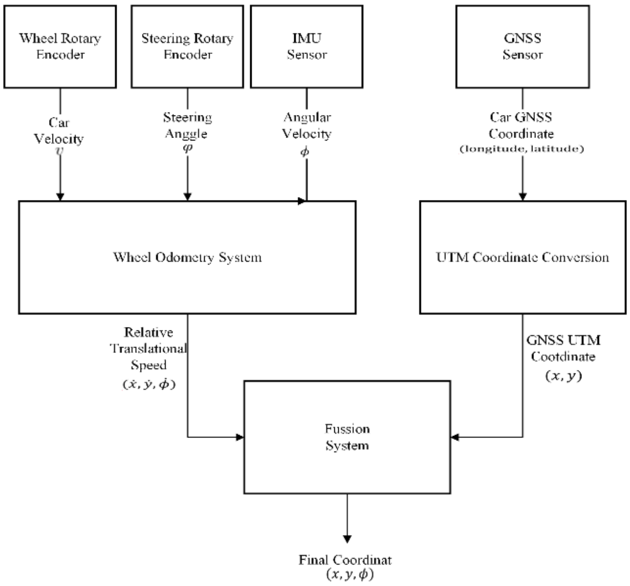

Fig 1. System Design.

The system utilizes a rotary encoder to measure the velocity of the car and the steering angle. Additionally, the angular velocity is determined by an Inertial Measurement Unit (IMU). These three inputs collectively generate relative translational velocity. This information is further integrated with GNSS UTM coordinates, obtained through the transformation of data from the GNSS sensor. The output of this fusion process provides the x, y, and ϕ coordinates of the car's position.

B. *Wheel Odometry System*

Within the framework of this system, a pair of rotary encoders plays a pivotal role. These encoders are intricately connected to each wheel of the vehicle, serving the fundamental purpose of gauging the speed of the car wheels. This real-time speed data, derived from the rotary encoders, becomes instrumental in executing forward kinematic calculations. These calculations, in turn, contribute to the determination of the car's precise position. By harnessing the rotational information from both wheels through the rotary encoders, the system can effectively conduct calculations that translate wheel speed into valuable position data, providing a comprehensive understanding of the vehicle's spatial coordinates.

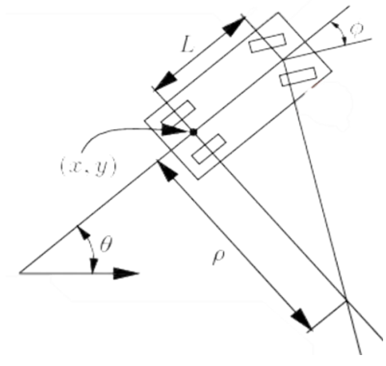

Fig. 2 Kinematic of Ackerman Steering

By using the Ackerman kinematic model, the relative position of the car to the starting point can be calculated using the following equation.

$$\dot{x} = s\cos\theta \tag{1}$$
$$\dot{y} = s\sin\theta \tag{2}$$
$$\dot{\theta} = \frac{s}{L}\tan\varphi \tag{3}$$

The car's relative coordinates (x, y) and heading (θ) in the Cartesian system are calculated based on the speed (s), axle-axle length (L), and steering angle (φ). This method provides fast data updates but accumulates errors over time due to wheel slippage and orientation sensor drifting. Additionally, the sysem resets to coordinate zero at startup. The theta angle significantly influences position outcomes, determined either through Ackerman steering kinematics or IMU angular velocity ($\dot{\phi}$). Given its precision, IMU angular velocity is favored over Ackerman steering, which introduces mechanical slip and increases error. Equations 1 and 2 substitute the IMU-derived theta for accuracy, obtained by integrating angular velocity ($\dot{\phi}$) readings from the IMU sensor.

$$\theta_{IMU} = \int\dot{\phi} \tag{4}$$
$$\dot{x} = s\cos\theta_{IMU} \tag{5}$$
$$\dot{y} = s\sin\theta_{IMU} \tag{6}$$

C. *GNSS System*

The fundamental operational concept of the GNSS involves receivers measuring the time-of-arrival of satellite signals and comparing it with the transmission time. This process facilitates the calculation of the signals' propagation time. The derived time propagations are then leveraged to gauge the distances from the GNSS receiver to the satellites, commonly referred to as range estimates. GNSS can simply be described as a navigation technology that is useful for showing the location of a place or object. In more detail, GNSS is an interconnected satellite-based navigation system. GNSS supports very accurate measurements of positioning, navigation, and time and also covers the entire world. Data from GNSS is presented in longitude and latitude coordinates so it needs to be converted first into Cartesian coordinate format using the following projection method.

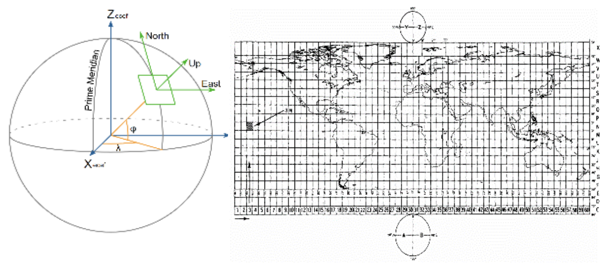

Fig. 2 UTM Conversion[4], [5]

UTM (Universal Transverse Mercator) coordinate system is basically a geographical latitude longitude system that is expressed in a two-dimensional projection of the surface of the earth where the earth map is divided into 60 zones, with each of them separated by 6 degrees in longitude and the locations are expressed in terms of so-called easting and northing, this system is most used for military purposes, x is Easting and y is Northing [6]. The UTM system divides the Earth into zones, and each zone is six degrees of longitude wide, so the calculation of UTM zone (N) can be done using this equation.

$$N = \left\lfloor \frac{Long+180}{6} + 1 \right\rfloor \tag{7}$$

The value from latitude and longitude need to convert from degree to radian first before performing the equation, each UTM zone has a central meridian, which is the longitude at the center of that zone, The central meridian (CM) for the given zone is calculated as

$$CM = 6(N-1) - 180 \tag{8}$$

After obtaining the CM value to obtain the x and y coordinate values, the following calculations are carried out.

$$x = k_0 v(\lambda - CM) \tag{9}$$

$$y = k_0 \left(p - M + (\phi - \phi_0)\frac{v}{2}\right) \tag{10}$$

Where k0 is the specific ellipsoid used for the Earth's shape, the value of k0 of central scale factor is typically very close to 0.9996 [6].

D. Fusion System

In this research, the proposed fusion system uses Extended Kalman Filter (EKF) because it is more flexible than ordinary Kalman Filters, where EKF can handle non-linear systems. Here is the motion model of the car used in this fusion.

$$\dot{x} = s\cos\theta \tag{11}$$

$$\dot{y} = s\sin\theta \tag{12}$$

$$\dot{\phi} = \omega \tag{13}$$

Based on the motion model the state model is:

$$X_{t+1} = f(X_t, U_t) = F_{X_t} + B_{U_t} \tag{14}$$

In this context, Δt represents the sampling time of the EKF program, which is set at 0.02 seconds.

With the matrix of F and B is:

$$F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} \cos(\phi)\Delta t & 0 \\ \sin(\phi)\Delta t & 0 \\ 0 & \Delta t \\ 1 & 0 \end{bmatrix} \tag{15}$$

So, the equation can be simplified to:

$$\begin{bmatrix} x' \\ y' \\ \omega' \\ v' \end{bmatrix} = f(X,u) = \begin{bmatrix} x + v\cos(\phi)\,\Delta t \\ y + v\sin(\phi)\,\Delta t \\ \phi + \omega\Delta t \\ v \end{bmatrix} \tag{16}$$

With the Jacobian matrix:

$$J_f = \begin{bmatrix} 1 & 0 & -v\sin(\phi)\Delta t & \cos(\phi)\Delta t \\ 0 & 1 & v\cos(\phi)\Delta t & \sin(\phi)\Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{17}$$

Information on the coordinates of the car's x, y position is obtained from GPS so that the observation model of the system is:

$$z_t = g(X_t) = HX_t \tag{18}$$

With :

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{19}$$

So, Observation function state:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = g(X) = \begin{bmatrix} x \\ y \end{bmatrix} \tag{20}$$

The final stage of this process is the prediction and update process where this process is formulated as follows.
Predict process:

$$x_{Pred} = Fx_t + Bu_t \tag{21}$$

$$P_{Pred} = J_f P_t J_f^T + Q \tag{22}$$

Update process:

$$z_{Pred} = Hx_{Pred} \tag{23}$$

$$y = z - z_{Pred} \tag{24}$$

$$S = J_g P_{Pred} . J_g^T + R \tag{25}$$

$$K = P_{Pred} . J_g^T S^{-1} \tag{26}$$

$$x_{t+1} = x_{Pred} + K_y \tag{27}$$

$$P_{t+1} = (I - KJ_g)P_{Pred} \tag{28}$$

The process of updating in the Extended Kalman Filter essentially iterates between prediction and correction steps as new sensor measurements become available. This iterative process helps in continuously refining the estimate of the system's state, both the system dynamics and the sensor measurements. The non-linearities are handled through linearization, making it a powerful tool for state estimation in non-linear systems like autonomous car localization, The EKF system from this research was built using the Robot Operating System (ROS) framework using the Robot Localization package [7].

III. EXPERIMENTAL

In this research, several data that need to be taken and analyzed include position data resulting from odometry and IMU data processing, as well as position data resulting from processing from the GNSS and IMU systems. each of these data will be recorded on a computer and presented in the form of a visual trajectory while the car is running. The car used in this experiment is the iCar, which is an autonomous electric car research platform at ITS.

Fig 4. Testing System at ICAR ITS

This iCar car has a GNSS antenna located on top of the car and a rotary encoder connected to the two rear wheels of the car as well as a Gyroscope sensor located on the car dashboard. The information is gathered by driving the car over a total distance of 807.71 meters.

A. Performance Comparation of Each System

Subsequently, the trajectory of each system, namely Odometry, GNSS, and Fusion Results, is documented and compared with the ground truth data. For this experiment, the ground truth is established using ublox ZED-F9P GNSS with the RTK system, which boasts centimeter-level accuracy as indicated in its datasheet. The resulting data is visualized in figure 5 below.

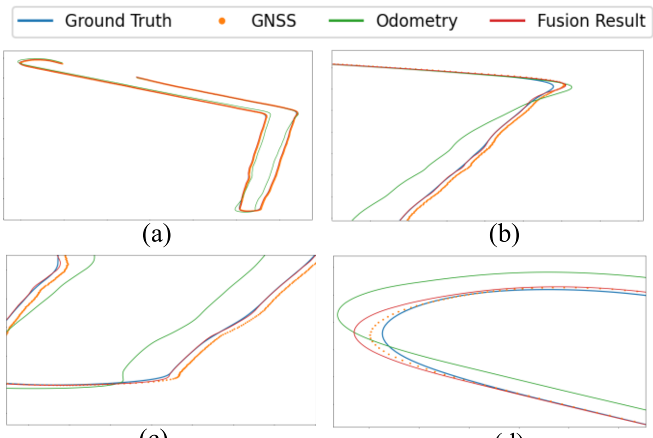

Fig 5. (a) Overall Trajectory, (b) 1st Highlight Trajectory (c)2nd Highlight Trajectory, (d) 3rd Highlight Trajectory

From the results obtained, in the first 100 meters before the turn occurs, all the data, both odometry, GNSS and ground truth data, are relatively the same, but when entering the first turn, the odometry data tends to widen or deviate further, this is due to slippage on the wheels. making calculations wrong where the wheels are detected rotating, but the position of the car does not change much. The error over time of each data is shown in Figure 6 below.

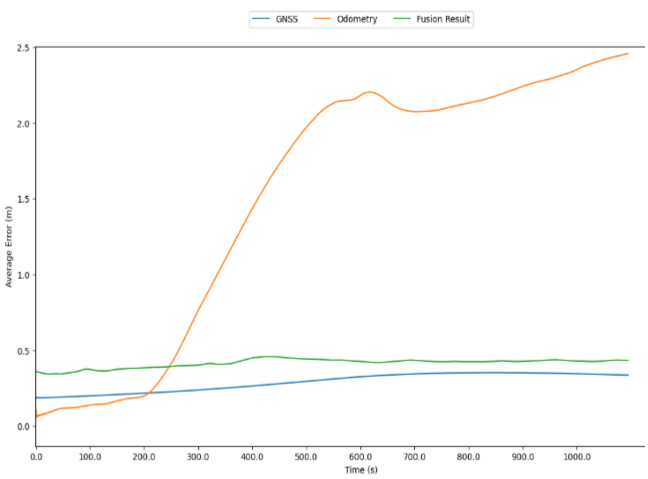

Fig 6. Error Overtime Graphic

Over the course of time, the data obtained from odometry exhibits a gradual increase, introducing cumulative inaccuracies in the integration of incremental measurements. This phenomenon may be attributed to various factors, including the effects of systematic biases that become more pronounced over time. data from GNSS has the smallest error, but keep in mind that this data has a very low update rate (10 Hz), which is not enough to use as a control reference for autonomous car systems. The continuous accumulation of errors in the odometry data poses a challenge to accurate localization, as it results in a growing disparity between estimated and actual positions. Based on the data that has been obtained, each error value can be calculated for the ground truth data using the following formula

$$ME = \frac{\sum_{i=1}^{n}(X_i - T_i)}{n} \quad (29)$$

The average error value (ME) is the average error value of the x and y coordinates so that the quality of each data, both odometry, GNSS and fusion results, is characterized as follows. The highest and lowest errors from each system were documented for the purpose of evaluating and contrasting the performance of each individual system.

TABLE I. SYSTEM PERFORMANCE DATA IN 800M DISTANCE

Data From	ME	Max Error	Min Error	Update Rate
Odometry	1.52 m	2.46 m	0.06 m	50 Hz
GNSS	0.31 m	0.50 m	0.15 m	10 Hz
Fusion Result	0.43 m	1.71 m	0.02 m	50 Hz

From the gathered data, it's apparent that odometry data presents the highest average error, while concurrently displaying the lowest minimum error value. This corresponds to the nature of odometry data, which exhibits commendable accuracy in the short term. However, it's important to note that over time, the error value tends to accumulate, resulting in an incremental increase. In comparison, GNSS data maintains relative stability in its error value and boasts the smallest average error. It's worth mentioning that the update rate of GNSS data is notably low, specifically at 10 Hz. On the other hand, fusion data can sustain errors at a level comparable to GNSS data but stands out due to a significantly higher update frequency, being five times faster than that of GNSS data.

B. Experimental of EKF Parameter

Tuning an Extended Kalman Filter (EKF) for localization using GNSS (Global Navigation Satellite System) and odometry involves adjusting parameters to achieve optimal performance. The EKF is a recursive Bayesian filter that combines measurements from multiple sensors to estimate the state of a dynamic system. In the context of localization, it fuses information from GNSS and odometry to estimate the position and orientation of a vehicle in this research, the parameters subjected to fine-tuning are Process Noise Covariance (Q) and Measurement Noise Covariance (R). Regarding Q, increasing the process noise covariance enables better adaptation to dynamic changes, but excessive values may lead to instability. Concerning R, modifications to this matrix impact the weight assigned to sensor measurements by the filter. To determine the most optimized parameters, multiple configurations were experimented with in this study. The parameters that are used in every experiment are presented in Table II, and the result is presented in Table III.

TABLE II. EXPERIMENT PARMETER

Experiment Number	Covariance	
	R	Q
1	[1.50, 1.50]	[0.10, 0.10] [1.00, 1.00]
2	[1.50, 1.50]	[0.20, 0.20] [1.00, 1.00]
3	[1.50, 1.50]	[0.50, 0.50] [1.00, 1.00]
4	[1.50, 1.50]	[0.01, 0.01] [1.00, 1.00]

TABLE III. RESULT EXPERIMENT PARMETER

Experiment Number	ME	Max Error	Min Error
1	0.43 m	1.71 m	0.020 m
2	0.38 m	1.64 m	0.015 m
3	0.40 m	1.75 m	0.012 m
4	0.53 m	2.20 m	0.024 m

In the context of this experiment, the variable subjected to adjustment is the Process Noise Covariance (Q). The significance of this parameter lies in its influence on the sensitivity of the system to changes in data, particularly in the GNSS measurements. A higher value of Process Noise Covariance renders the system more responsive to alterations in GNSS data, amplifying its sensitivity to external changes. Conversely, a smaller value of Q results in reduced sensitivity to variations in GNSS data, indicating a heightened reliance on odometry data. In essence, manipulating the Process Noise Covariance provides a means to fine-tune the system's responsiveness to different sources of information, offering a nuanced control over its sensitivity to GNSS and odometry data. This experimental exploration of Q serves as a pivotal step in optimizing the system's robustness and adaptability to diverse environmental conditions and sensor inputs.

IV. CONCUSION

The analysis of results, as presented in Table I, reveals a notable trend. Specifically, it becomes evident that the data derived from GNSS exhibits the smallest average error value, registering at a mere 0.31 meters. However, a nuanced comparison with the fusion results, indicating an average error of 0.43 meters, underscores a slight disparity. It is essential to underscore a crucial aspect - the data from GNSS is characterized by an exceptionally low update rate. This limited frequency of updates renders it impractical for deployment as a control reference in the context of autonomous vehicles. Consequently, while the standalone GNSS data showcases superior accuracy, its inadequate update speed necessitates an alternative approach. In light of this, the fusion data emerges as a more reliable option, presenting a pragmatic compromise that balances precision with a sufficiently high update rate, rendering it a viable and dependable choice for deployment in autonomous vehicle systems. Adjusting the performance of sensor fusion involves tweaking the Process Noise Covariance (Q), a critical parameter. However, it's important to note that changing Q can result in different effects across various systems. Each system's distinct features and sensors play a role in determining the most suitable value for Q. Therefore, a tailored and system-specific approach to tuning Q is crucial. This adaptability ensures that the fusion algorithm is finely tuned to the specific characteristics of each system, enhancing its effectiveness in combining different sensor inputs for accurate localization in autonomous applications.

V. ACNOWLEDGMENT

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